

Linux Task 03 Nimitha: Process Management, System Monitoring & Basic Shell Scripting

Objective The purpose of this task is to understand how Linux manages processes, monitors system resources, and automates tasks using shell scripts. These are essential skills for Linux Administrators, SOC Analysts, and Cyber Security Professionals. Part A: Process Monitoring

Part A: Process Monitoring

- ps
- ps aux
- top
- htop

1. What is a Process?

A process is a program that is currently running on the system. Every application or command executed in Linux runs as a process

2. What is a PID?

PID stands for Process ID. It is a unique number assigned to every running Process.

3. Which process is consuming the most CPU?

The process consuming the most CPU is redis.

- PID: 123
- Process Name: redis
- CPU Usage: approximately 0.9%

4. Which process is consuming the most Memory?

the process using the most memory is root process with PID 123.

- PID: 123
- Memory Usage (RES): 31140 KB
- %MEM: approximately 0.3%

Part B: Process Management

Item -- Value

Process Name --sleep

PID Found-- 2134

Command Used -- kill 2134

Result -- Process terminated successfully

Part C: System Monitoring

- free-h
- df-h
- uptime
- uname -a

Information Output

Total RAM -- 6.4 GiB

Available RAM-- 7.8 GiB

Disk Usage -- /dev/sdd: 1017G total, 2.3G used, 914G
available (1% used)

System Uptime 3:08 (2 hours 2 minutes)

Kernel Version Linux LAPTOP 6.6.114.2-microsoft

Part D: Service Monitoring

- systemctl status ssh

- `systemctl status NetworkManager`

1. What is a Service?

A service is a background program that runs continuously to provide specific functions such as networking, logging, or web hosting.

2. Why are services important?

Services keep essential system functions running automatically. They ensure that applications and system components work properly.

3. How can a stopped service affect a system?

If an important service stops, related functionality may fail. For example, if the network service stops, internet connectivity may be lost.